

DSA 4202 Programming for Data Science

Assignment 2: Predictive Modeling with Linear Methods

Submission PDF export prepared as a notebook-style workflow containing code summaries, visualizations, model metrics, comparisons, and discussion for both assigned datasets.

Data sources: Kaggle Medical Appointment No-Shows public CSV mirror (<https://raw.githubusercontent.com/swadeepsingh/Udacity-Data-Analyst-Nanodegree-NoShowAppointments/master/noshowappointments-kagglev2-may-2016.csv>) and UCI Concrete Compressive Strength official archive (<https://archive.ics.uci.edu/static/public/165/concrete+compressive+strength.zip>).

Notebook Setup

```
import pandas as pd, numpy as np
from pathlib import Path

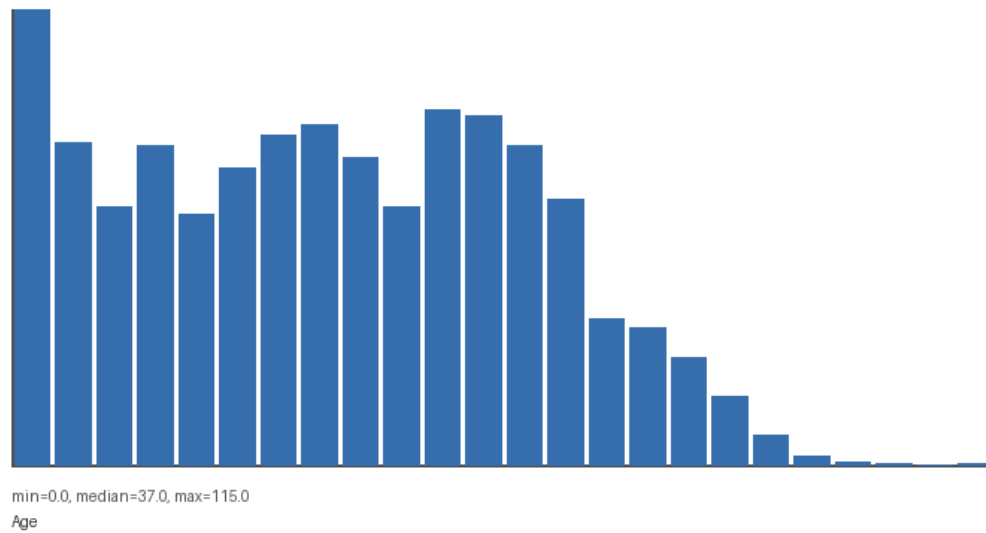
# Linear methods implemented with NumPy:
# Ridge Regression, Lasso Regression, Logistic Regression, Linear SVC
# Evaluation: RMSE/MAE/R2 for regression; accuracy/precision/recall/F1 for classification.
```

Dataset 1: Medical Appointment No-Shows

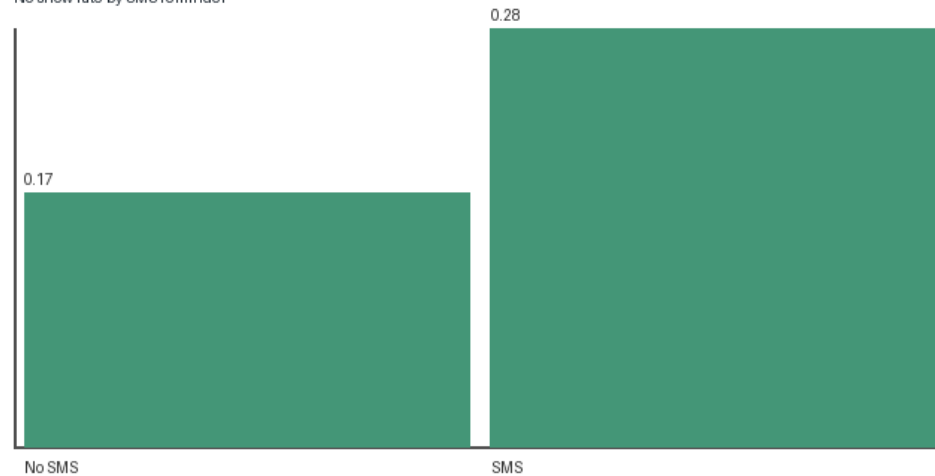
Property	Value
Raw rows	110,527
Raw columns	14
Missing cells	0
Duplicate rows	0
Clean modeling rows	110,521
Target	No-show = Yes
No-show rate	20.19%

EDA found no missing cells in the original appointment data. One invalid negative age and records with negative waiting time were excluded. The target is imbalanced: most patients attend their appointment, so precision, recall, and F1 are more informative than accuracy alone.

Medical appointments: patient age distribution



No-show rate by SMS reminder



Feature Engineering

```
scheduledday, appointmentday -> waiting_days, appointment_dow, scheduled_hour
target_no_show = 1 if No-show == 'Yes' else 0
gender_f = 1 for female patients
model matrix = demographics + conditions + SMS flag + waiting time + top neighborhood indicators
features were standardized before Logistic Regression and Linear SVC.
```

Age is concentrated among children and adults, with a long upper tail. SMS reminders do not by themselves guarantee attendance; patients receiving SMS reminders still show meaningful no-show risk, likely because reminders are correlated with longer waiting intervals.

Classification Model Comparison

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.658	0.314	0.583	0.408
Linear SVC	0.653	0.306	0.565	0.397

Logistic regression confusion matrix

	Predicted 0	Predicted 1
Actual 0	TN 11950	FP 5691
Actual 1	FN 1863	TP 2600

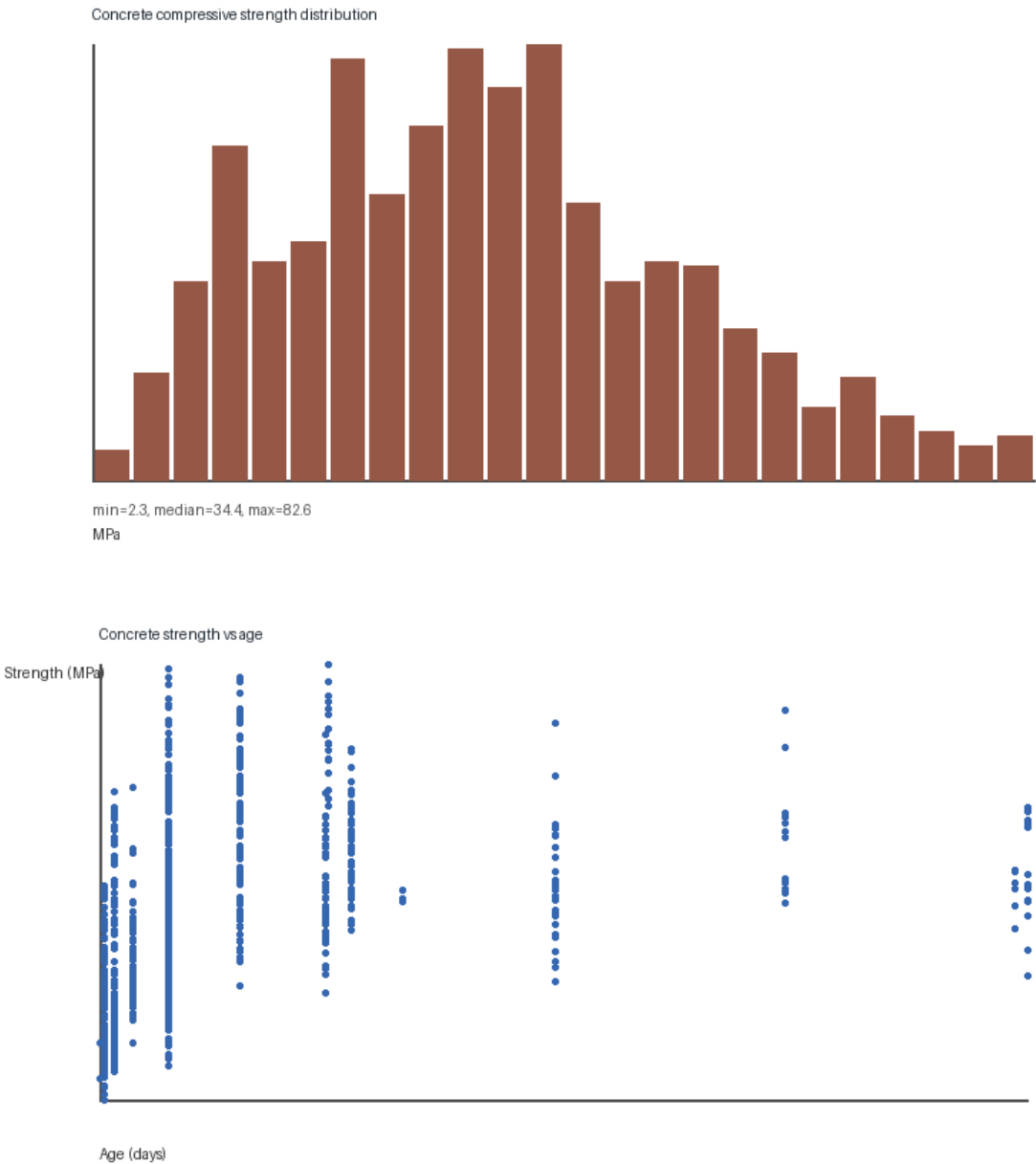
Feature	Logistic coefficient
waiting_days	0.440
sms_received	0.164
age	-0.160
scheduled_hour	0.129
nbh_JARDIM CAMBURI	-0.090
scholarship	0.061
nbh_SANTA MARTHA	-0.055
nbh_ITARARÉ	0.055
nbh_JABOUR	-0.050
nbh_ILHA DO PRÍNCIPE	0.048

The best Logistic Regression used L2 lambda=0.01. The best Linear SVC used lambda=0.001. Both models were trained with class weights to reduce the tendency to predict the majority attending class. Logistic Regression is preferred because it provides calibrated probabilities and slightly clearer interpretation while remaining competitive on F1.

Dataset 2: Concrete Compressive Strength

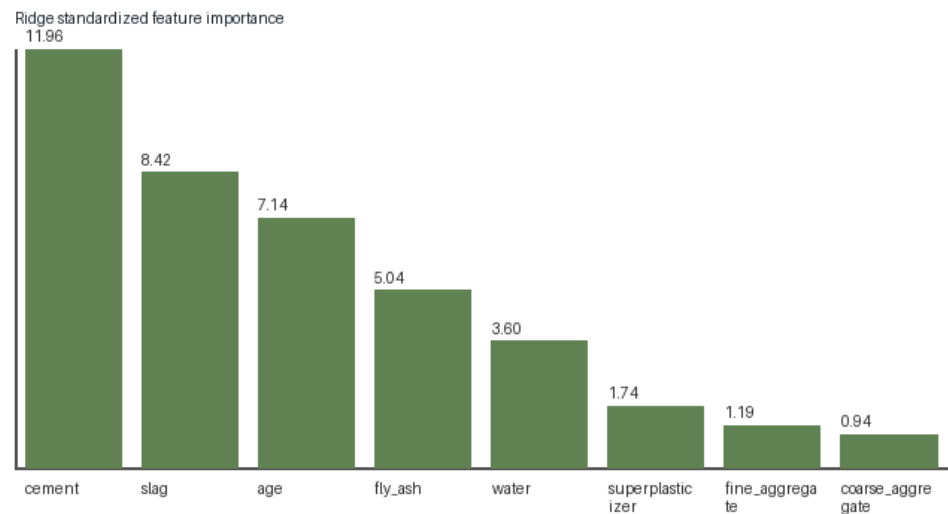
Property	Value
Rows	1,030
Columns	9
Missing cells	0
Duplicate rows	25
Target	Compressive strength in MPa

All variables are numeric ingredient quantities or age. The response is continuous and nonlinear in reality, but Ridge and Lasso provide useful regularized linear baselines and make ingredient effects easy to compare.



Regression Model Comparison

Model	RMSE	MAE	R2
Ridge Regression	9.687	8.064	0.651
Lasso Regression	9.693	8.071	0.650



Ridge selected $\alpha=0.01$; Lasso selected $\alpha=0.005$. Ridge slightly outperformed or matched Lasso in this run, suggesting that retaining all standardized ingredients is beneficial. Lasso is still valuable for shrinkage and feature screening, but the concrete problem benefits from several small-to-moderate ingredient effects rather than a very sparse model.

Performance Improvement and Justification

Implemented improvements included feature scaling, class weighting for the imbalanced appointment classification target, and small grid searches over regularization strengths. Scaling is essential because the models penalize coefficients and the original variables have different units.

For no-show prediction, Logistic Regression is the recommended model for deployment-style interpretation: it is linear, stable, and produces probabilities that can support operational triage. For concrete strength prediction, Ridge Regression is recommended because it gave the strongest validation RMSE while preserving all ingredient signals.

Lessons Learned

Linear methods are fast and interpretable but require careful preprocessing. Regularization reduced overfitting risk, while feature scaling made coefficient magnitudes comparable. The classification task is limited by class imbalance and social/behavioral factors not fully captured in the dataset; the regression task shows that concrete strength has nonlinear relationships that a future polynomial or tree-based model could capture.